

Supplementary Material

Analytical Modeling of a Soft Pneu-Net Actuator Subjected to Planar Tip Contact

Sachin, Zhongkui Wang, and Shinichi Hirai

1 Modeling of Air Chambers

The modeling of air chambers as a flat membrane is discussed in this section. The detailed derivation of the governing equations for an initially flat membrane is discussed in detail in the subsequent subsections. The deformed profile of air chambers is described in next subsection.

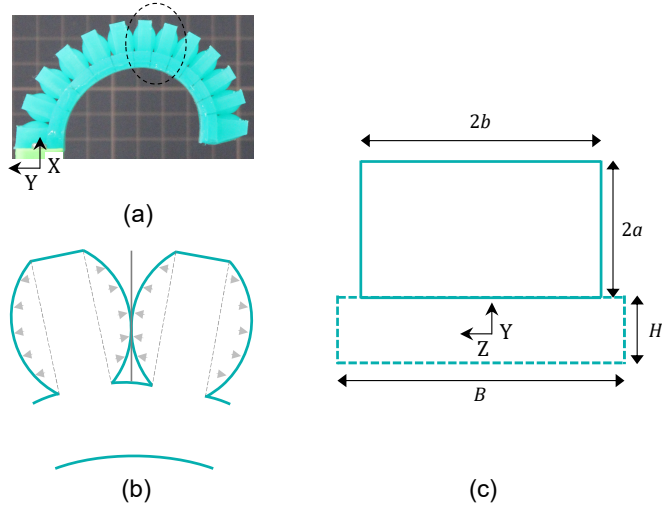


Figure 1: Geometry of air chambers: (a) photo snapshot of a soft pneu-net actuator subjected to air pressure in free space deformation state, (b) enlarged sketch for the deformation of adjacent air chambers, and (c) sectional view at the edge of air chamber along YZ plane, it represents the undeformed side wall. In (c) the solid boundary represents the air chambers side walls (membrane) and the dotted boundary represented the actuator cover (beam). The labels are: $2a$ width of membrane, $2b$ length of membrane, H thickness of actuator cover, and B width of actuator cover.

1.1 Finite Deformation

The pneu-net soft actuator consists of several serially connected air chambers. In the geometric design of air chambers, the thickness of the chamber walls facing the adjacent chambers (in YZ plane) is lower than that of the remaining walls. When inflated in free space state, the actuator attains the deformed shape, as shown in Fig. 1(a). A schematic for the adjacent chambers in contact is shown in Fig. 1(b). The adjacent side walls are modeled as flat membranes. The undeformed membrane is flat and it is assumed to be rigidly clamped at the edges. A schematic for the undeformed membrane with sides $2a$ and $2b$ is shown in Fig. 1(c). For thin films or membranes, the bending stiffness is negligible. In this study, the inflated membrane is assumed to have a circular geometry, as shown in Fig. 1(b). A detailed schematic for the deformation of adjacent air chambers

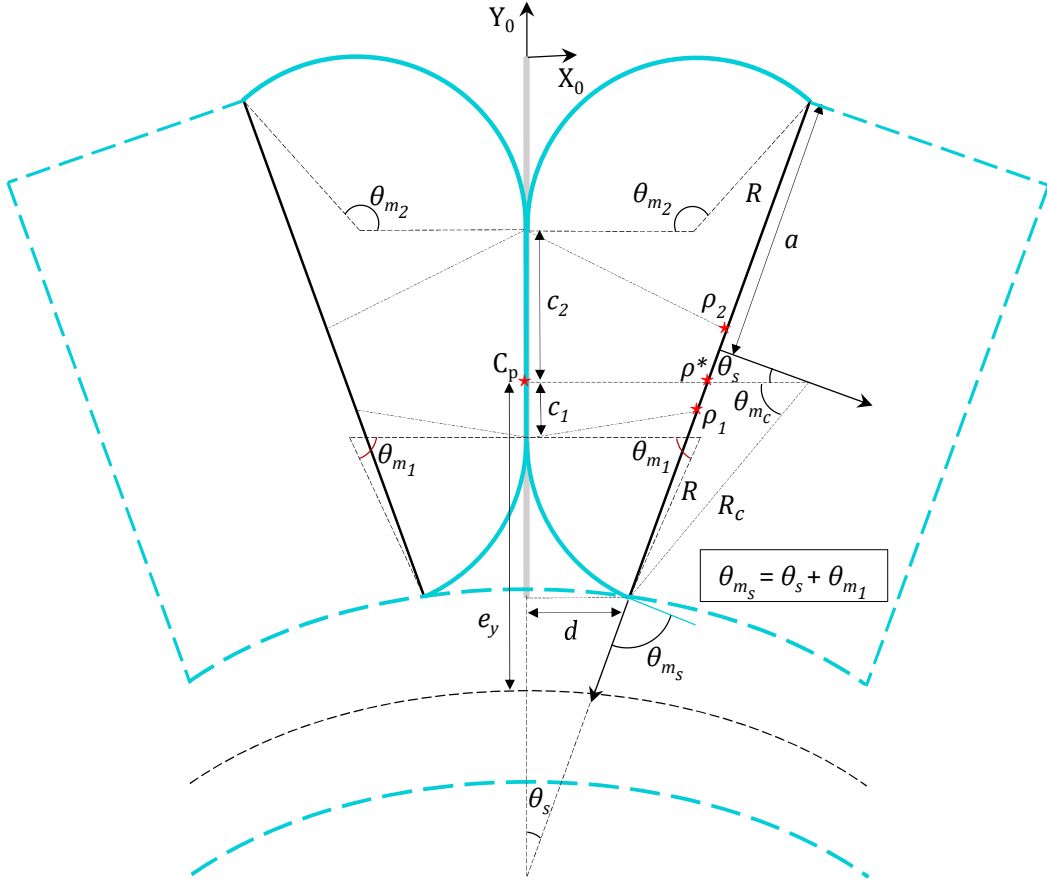
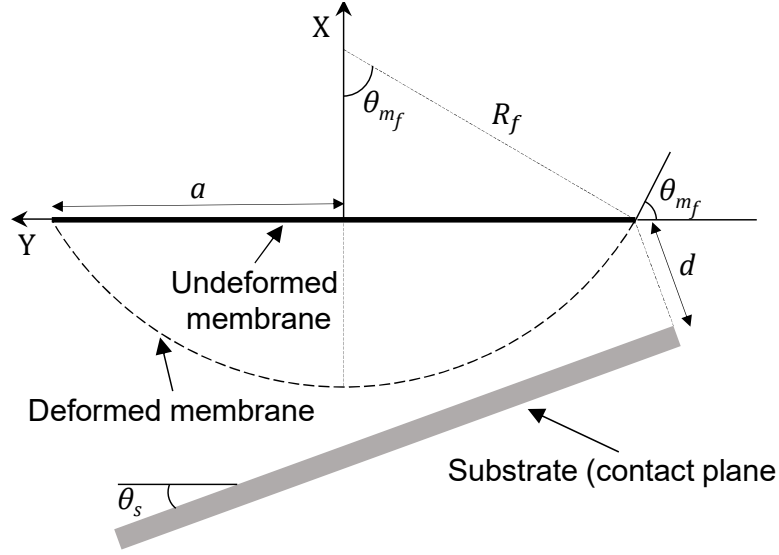


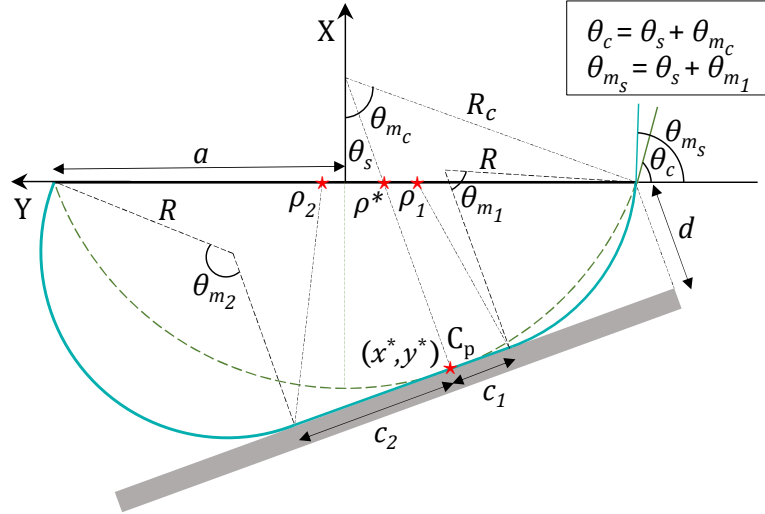
Figure 2: Schematic for the deformed geometry of air chambers. The solid aquamarine color lines represents the deformed profile of membranes and the dashed line represents the air chambers and actuator cover. The labels are: $2a$ width of membrane, d offset between membrane and substrate at the interior, θ_s substrate inclination angle or half the inclination angle between adjacent air chambers, ρ^* undeformed location of contact initiation point C_p , ρ_1 undeformed location of interior contact edge, ρ_2 undeformed location of exterior contact edge, R radius of curvature, R_c radius of curvature at contact initiation, θ_{m_2} exterior angle, θ_{m_1} interior angle, θ_{m_c} angle subtended by interior arc at contact initiation, θ_{m_s} slope at interior clamp edge, c_1 and c_2 contact widths, and e_y eccentricity.

in contact is shown in Fig. 2. The derivation of various geometric parameters mentioned in the schematic is discussed in the later subsections.

The deformation of membranes is assumed to be symmetric about contact plane Y_0Z_0 . Initially, in undeformed state, the contact plane is parallel to the membranes, but once the actuator is deformed, it becomes inclined to the membranes. Towards the interior of deformed air chambers, the membrane is at a distance d from the contact plane (later in the derivation, the contact plane is assumed to be a rigid substrate), that is inclined to the undeformed membrane surface at an angle θ_s , as shown in Fig. 3(a). The deformed profile of an initially flat membrane in contact with an inclined substrate is shown in Fig. 3(b). We refer the deformation to cartesian coordinates that are defined along the undeformed membrane surface. We defined the surface of deformed membrane that is not in contact with the substrate as free surface, and it is circular in geometry [1]. The dotted curve in Fig. 3(b), denotes the deformed profile of membrane at contact initiation. The contact begins at point $C_p(x^*, y^*)$ and its location is dependent on parameters d and θ_s . The contact width, unlike that in contact with parallel substrate, is not symmetric about the point of contact initiation and is defined by c_1 and c_2 from the point of contact initiation, along the substrate. The next subsection presented a discussion on the constitutive relations for a square membrane. The deformation of rectangular membrane is predicted from the deformation of square membrane using stretch mapping.



(a)



(b)

Figure 3: Deformed geometry of flat membrane: (a) before contact and (b) in contact with an inclined substrate. The dotted curve indicates the free deformation. The labels are: R_f radius of curvature of free membrane before contact and θ_{m_f} slope of free membrane at clamp edge before contact.

1.2 Constitutive Relations for Membrane

The free deformation of membrane is symmetric about X -axis, and two of the principal stretches at the center of membrane are assumed to be equal. The membrane is assumed to be isotropic, incompressible, and hyperelastic. The principal stretches are defined as $\lambda_1 = \lambda_2 = \lambda_m$, and from the incompressibility condition $\lambda_3 = \lambda_m^{-2}$. The strain energy density $U(I_1, I_2)$ is a function of the strain invariants I_1 and I_2 . The strain invariants are given by

$$I_1 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2, \quad I_2 = \lambda_1^{-2} + \lambda_2^{-2} + \lambda_3^{-2}. \quad (1)$$

For hyperelastic membranes, the stress resultants or line tensions in principal directions are defined by [2], [3]

$$\begin{aligned} T_1 &= 2t_m \lambda_3 (\lambda_1^2 - \lambda_3^2) (U_1 + \lambda_2^2 U_2), \\ T_2 &= 2t_m \lambda_3 (\lambda_2^2 - \lambda_3^2) (U_1 + \lambda_1^2 U_2), \end{aligned} \quad (2)$$

where T_1 and T_2 are line tensions in principal directions, t_m is membrane thickness, and U_i denotes $\partial U / \partial I_i$. We denote by p the pressure applied on the membrane and by κ_1 and κ_2 the principal curvatures of the deformed membrane surface. The equilibrium equation for the membrane is defined as [3]

$$\kappa_1 T_1 + \kappa_2 T_2 = p. \quad (3)$$

As an assumption for the deformation state or principal stretches defined above for a square membrane, the tensions $T_1 = T_2 = T_m$ and the principal curvatures $\kappa_1 = \kappa_2 = \kappa_m$. The equilibrium equation (3) can then be written as

$$T_m = \frac{p}{2\kappa_m}. \quad (4)$$

The equilibrium equations are independent of membrane material. We consider incompressible neo-Hookean material, and the strain energy density function U is defined as [4]

$$U = \frac{\mu}{2} (I_1 - 3), \quad (5)$$

where μ is the shear modulus. From (5), we obtain the partial differentials as $U_1 = \mu/2$ and $U_2 = 0$. Substituting the stretches and partial differentials into (2), the tension T_m can be written as

$$T_m = \mu t_m \left(1 - \frac{1}{\lambda_m^6} \right). \quad (6)$$

From (4) and (6), we get the final equilibrium equation as

$$\frac{p}{2\kappa_m} = \mu t_m \left(1 - \frac{1}{\lambda_m^6} \right). \quad (7)$$

Now, the curvature κ_m is the only unknown parameter and it is defined in terms of the principal stretch λ_m from deformed geometry of the membrane.

1.3 Free Inflation of Membrane

The membrane experiences free inflation before contact initiation. Before contact, the center of deformed membrane lies on the X -axis, and it makes an angle θ_{m_f} at the clamped edge, as shown in Fig. 3(a). The stretch in the deformed membrane is uniform and geometrically, it can be calculated as

$$\lambda_m = \frac{R_f \theta_{m_f}}{a}. \quad (8)$$

The radius of curvature R_f can be expressed as

$$R_f = \frac{a}{\sin \theta_{m_f}}. \quad (9)$$

From (8) and (9), the stretch can be written as

$$\lambda_m = \frac{\theta_{m_f}}{\sin \theta_{m_f}}. \quad (10)$$

The maximum deflection δ of the deformed membrane can be determined as

$$\delta = R_f (1 - \cos \theta_{m_f}) = a \tan \left(\frac{\theta_{m_f}}{2} \right). \quad (11)$$

The curvature κ_m is the inverse of R_f for circular deformed surface of the membrane. Substituting (9) and (10) into (7), we get the relation between θ_{m_f} and the applied pressure p , as

$$\sin \theta_{m_f} - \frac{\sin \theta_{m_f}^7}{\theta_{m_f}^6} = \frac{pa}{2\mu t_m}. \quad (12)$$

Equation (12) has a limit on the upper bound of pressure p . After a certain pressure value, the equation has no solution. However, for the deformation of the membrane in contact, the stretch λ_m and R_f are defined using different expressions. In that case, the upper bound on pressure is imposed by θ_{m_2} , with an upper limit π .

For a soft pneu-net actuator, we can define the minimum pressure p_{ct} required to initiate the contact between adjacent air chambers using (12). The minimum inflation of air chambers required to make contact with adjacent chambers is $\delta = h_x$, where h_x is half the gap between adjacent air chambers. By substituting $\delta = h_x$ into (11) and solving it for θ_{m_f} , we get

$$\theta_{m_f} = 2 \tan^{-1} \left(\frac{h_x}{a} \right). \quad (13)$$

By rearranging (12), we get the expression for pressure as

$$p_{ct} = \frac{2\mu t_m}{a} \left(\sin \theta_{m_f} - \frac{\sin \theta_{m_f}^7}{\theta_{m_f}^6} \right). \quad (14)$$

1.4 Contact Initiation

The geometry of the membrane in contact with an inclined substrate is shown in Fig. 3(b). The contact begins at point $C_p(x^*, y^*)$, on the deformed membrane profile. From the deformed geometry, the location of point C_p can be defined as

$$x^* = -R_c (\cos \theta_s - \cos \theta_c) \text{ and } y^* = R_c \sin \theta_s. \quad (15)$$

The slope θ_c can be calculated as

$$\frac{d}{a} = \csc \theta_c - \cos \theta_s \cot \theta_c - \sin \theta_s. \quad (16)$$

The undeformed location $(0, \rho^*)$ of contact initiation point $C_p(x^*, y^*)$ can be written as

$$\rho^* = a \frac{\theta_s}{\theta_c}. \quad (17)$$

The contact initiation point is assumed to be stationary and the contact widths are defined with respect to this point.

1.5 Inclined Contact Between Membrane and Substrate

First, we will investigate the left side (exterior) of contact initiation point and obtain the expression for contact width c_2 ; then, from the deformed geometry, we develop the correlation between c_1 and c_2 . If the contact between the membrane and substrate is assumed to be frictionless, then the membrane has a uniform stretch. The stretch at the contact edge can be defined either from the

free membrane λ_{out} or from the membrane in contact λ_{in} . For frictionless contact, the stretches λ_{in} and λ_{out} are equal. From deformed geometry, the stretch at the contact edge can be written as

$$\lambda_{in} = \frac{c_2}{\rho^* - \rho_2} \quad (18)$$

and

$$\lambda_{out} = \frac{R\theta_{m_2}}{a + \rho_2} . \quad (19)$$

The undeformed location ρ_2 of contact edge can be obtained by equating (18) and (19), as

$$\rho_2 = \frac{R\theta_{m_2}\rho^* - ac_2}{R\theta_{m_2} + c_2} . \quad (20)$$

The radius of curvature R can be obtained from the deformed geometry, as

$$R = \frac{(a + \rho^*) \cos \theta_s - c_2}{\sin \theta_{m_2}} . \quad (21)$$

The undeformed contact width ρ_2 can be expressed in terms of θ_{m_2} by substituting (21) into (20), as

$$\rho_2 = \frac{\theta_{m_2}\rho^* ((a + \rho^*) \cos \theta_s - c_2) - ac_2 \sin \theta_{m_2}}{\theta_{m_2} ((a + \rho^*) \cos \theta_s - c_2) - c_2 \sin \theta_{m_2}} . \quad (22)$$

From the deformed geometry of membrane, the relation between radius of curvature R and offset d can be expressed as

$$d = R(1 - \cos \theta_{m_2}) - 2a \sin \theta_s . \quad (23)$$

The contact width c_2 can be expressed in terms of θ_{m_2} by substituting (21) into (23), as

$$c_2 = (a + \rho^*) \cos \theta_s - (d + 2a \sin \theta_s) \cot \left(\frac{\theta_{m_2}}{2} \right) . \quad (24)$$

The radius of curvature R can be expressed in terms of θ_{m_2} by substituting (24) into (21) or from (23), as

$$R = \frac{d + 2a \sin \theta_s}{2 \sin^2 \left(\frac{\theta_{m_2}}{2} \right)} . \quad (25)$$

The stretch λ_m can be calculated by substituting (22) and (24) into (18), as

$$\lambda_m = \cos \theta_s + \frac{d + 2a \sin \theta_s}{2(a + \rho^*)} \left[\frac{\theta_{m_2}}{\sin^2 \left(\frac{\theta_{m_2}}{2} \right)} - 2 \cot \left(\frac{\theta_{m_2}}{2} \right) \right] . \quad (26)$$

The tension T_m in free-standing portion of the membrane can be obtained by substituting (25) into (4), as

$$T_m = \frac{p}{2} \frac{d + 2a \sin \theta_s}{2 \sin^2 \left(\frac{\theta_{m_2}}{2} \right)} . \quad (27)$$

The equilibrium equation for the deformed membrane is obtained by substituting (25) and (26) into (7) and in the final form, it can be written as

$$1 - \lambda_m^{-6} = \frac{p(d + 2a \sin \theta_s)}{4\mu t_m \sin^2 \left(\frac{\theta_{m_2}}{2} \right)}. \quad (28)$$

We can obtain θ_{m_2} by solving (28). And by substituting it into (24) and (26), we can get c_2 and λ_m , respectively. Now, c_1 and the contact width along Z direction c_z are the unknown parameters. The approach to calculate the contact widths is discussed in the next subsection.

1.6 Contact Widths

The free standing portions of the membrane have a uniform radius of curvature, and using this argument, we can define the correlation between c_1 , θ_{m_1} , and R . From the deformed geometry, the clamp angle θ_{m_1} can be expressed in terms of radius R as

$$\theta_{m_1} = \cos^{-1} \left(1 - \frac{d}{R} \right). \quad (29)$$

The contact width on the right half (interior) can be obtained using the expression

$$c_1 = 2a\lambda_m + R(\theta_{m_1} + \theta_{m_2}) - c_2. \quad (30)$$

The maximum contact width along the Z direction exists along the point of contact initiation, and it can be obtained by equating the deformed and undeformed lengths along the Z direction; in the final form, it can be written as

$$c_z = (c_1 + c_2) \frac{b(a - \rho^*)}{a^2}. \quad (31)$$

In the equations derived above, if the substrate inclination angle θ_s will be set equal to zero then, we get the parallel contact model, and in that case, $\theta_{m_1} = \theta_{m_2}$ and $c_1 = c_2$.

Finally, by using (25), (28), and (29), we can obtain the deformed profile of the membrane. And, from (24), (30), and (31), we can calculate the contact widths to predict the contact force.

2 Finite Element Simulations for Pneu-net Actuator

Finite element (FE) simulations were performed for free space and block force configurations of the actuators. The free space deformation of the actuators, as predicted by the FE model, is shown in Fig. 4, and the block force deformation is shown in Fig. 5.

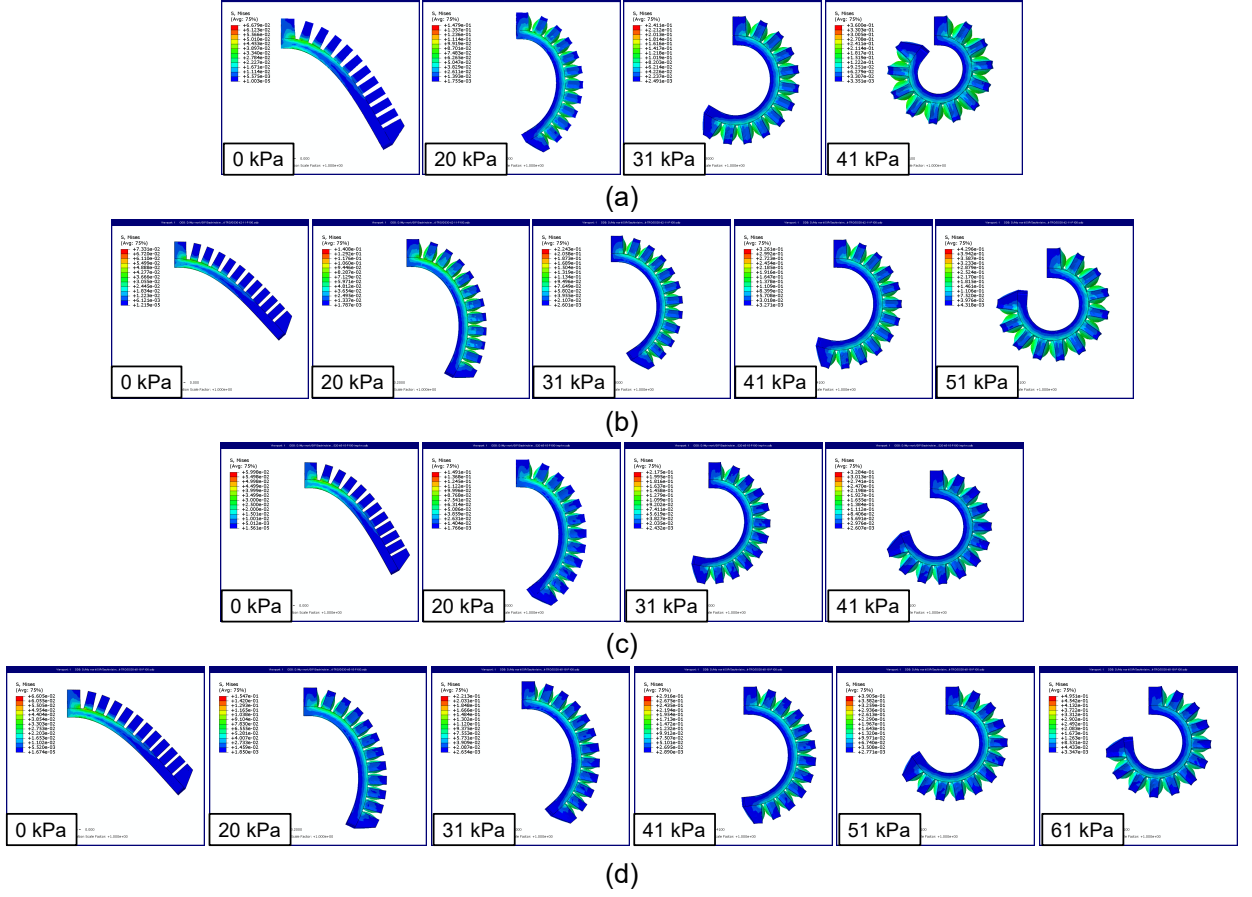


Figure 4: FE simulated free space deformation of (a) type A, (b) type B, (c) type C, and (d) type D actuators.

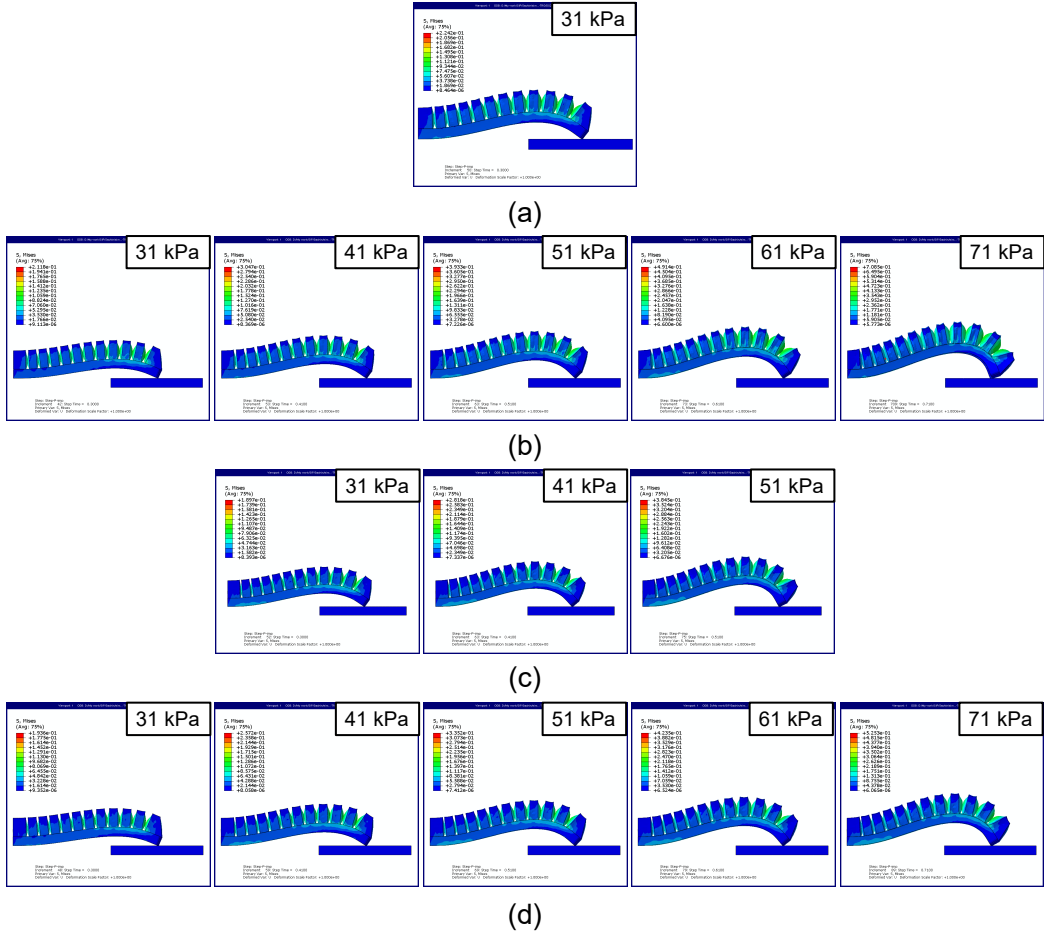


Figure 5: FE simulated block force deformation of (a) type A, (b) type B, (c) type C, and (d) type D actuators.

3 Block Force Results for Contact at Different Heights

The analytical computations and experiments were performed for different contact heights, at the tip. The experiments were performed for four different offsets: 10 mm, 20 mm, -10 mm, and -20 mm. A schematic for the experiments is shown in Fig. 6. The experiments and analytical simulations were performed for four different actuators. The force results are presented in Fig. 7. The force values estimated by the analytical model are close to the experimental force values. However, the theoretical model cannot predict the force value for negative offset after a certain air pressure magnitude. This happens due to the solver. The solver gives a solution that satisfies the boundary conditions but it is practically infeasible.

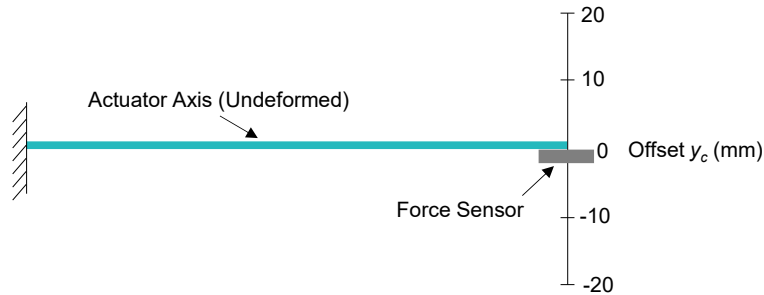


Figure 6: Schematic for the force measuring experiment for contact at different heights.

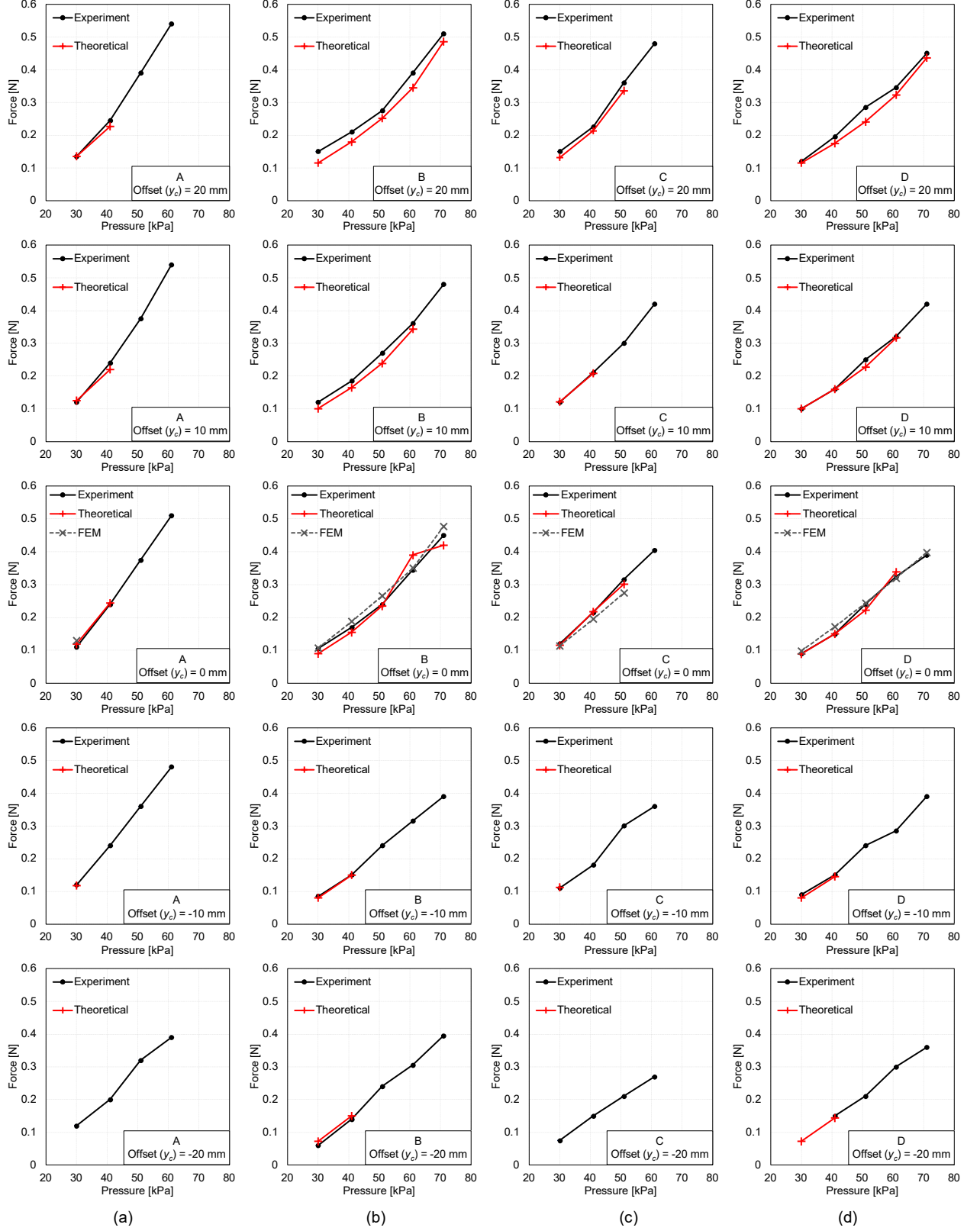


Figure 7: Theoretical and experimental block force plots of (a) type A, (b) type B, (c) type C, and (d) type D actuators, at different offset heights y_c .

4 Block Force Deformation of Pneu-net Actuators

The experimental, theoretical, and FE simulated block force deformation plots of the actuators are presented in Fig. 8. The analytical model overestimates the deformation, whereas the FE simulated deformation is close to the experimental deformation. This occurs because of the additional stiffness provided by the air chambers to the bottom cover of actuators. In the analytical model, this bending stiffness is not considered. This is a limitation of the proposed model.

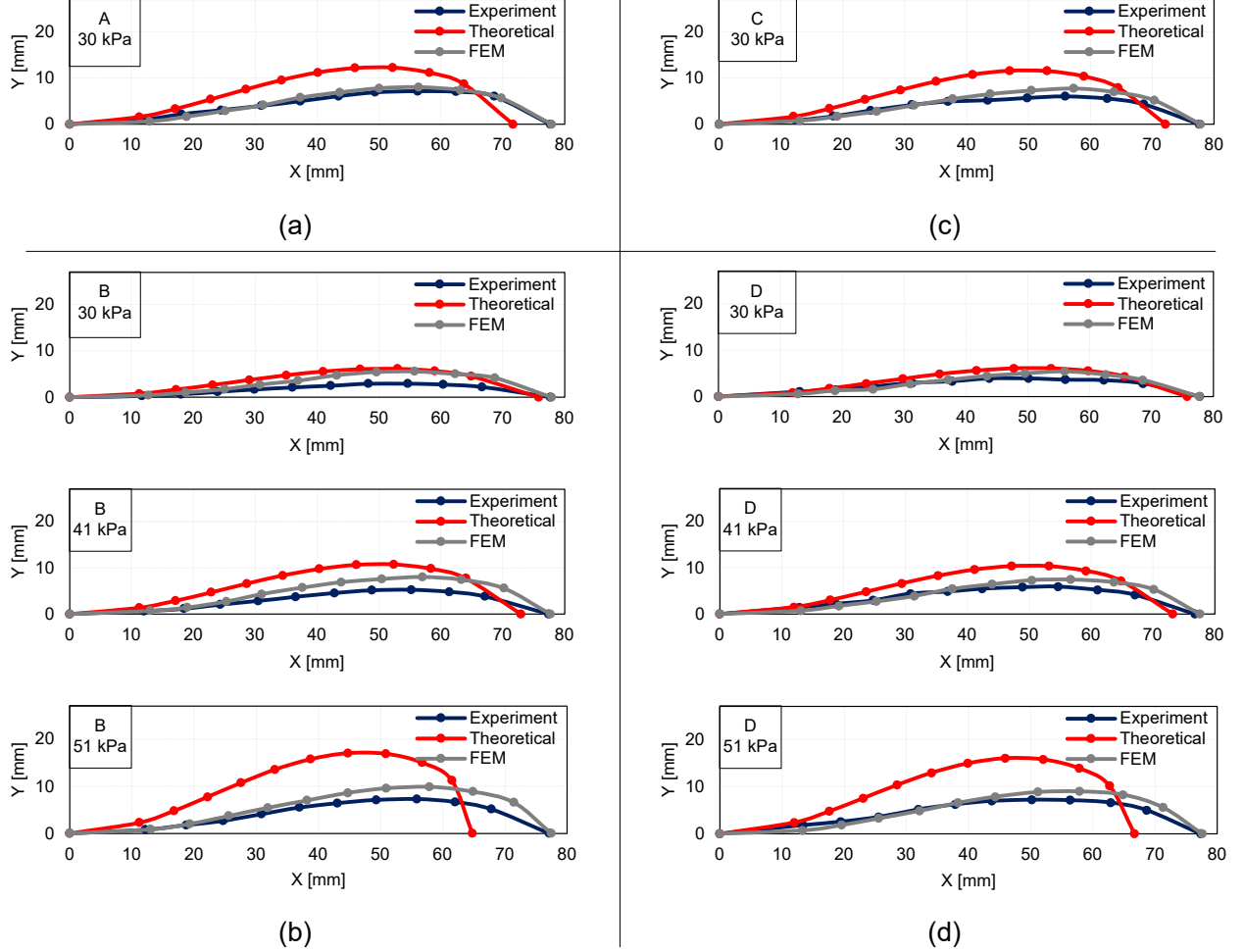


Figure 8: Block force deformation plots of (a) type A, (b) type B, (c) type C, and (d) type D actuators.

5 Pneu-net Actuator Having Nonuniform Geometry

The free space deformation of a pneu-net actuator with a nonuniform geometry is shown in Fig. 9. The actuators have denser chambers toward the tip.

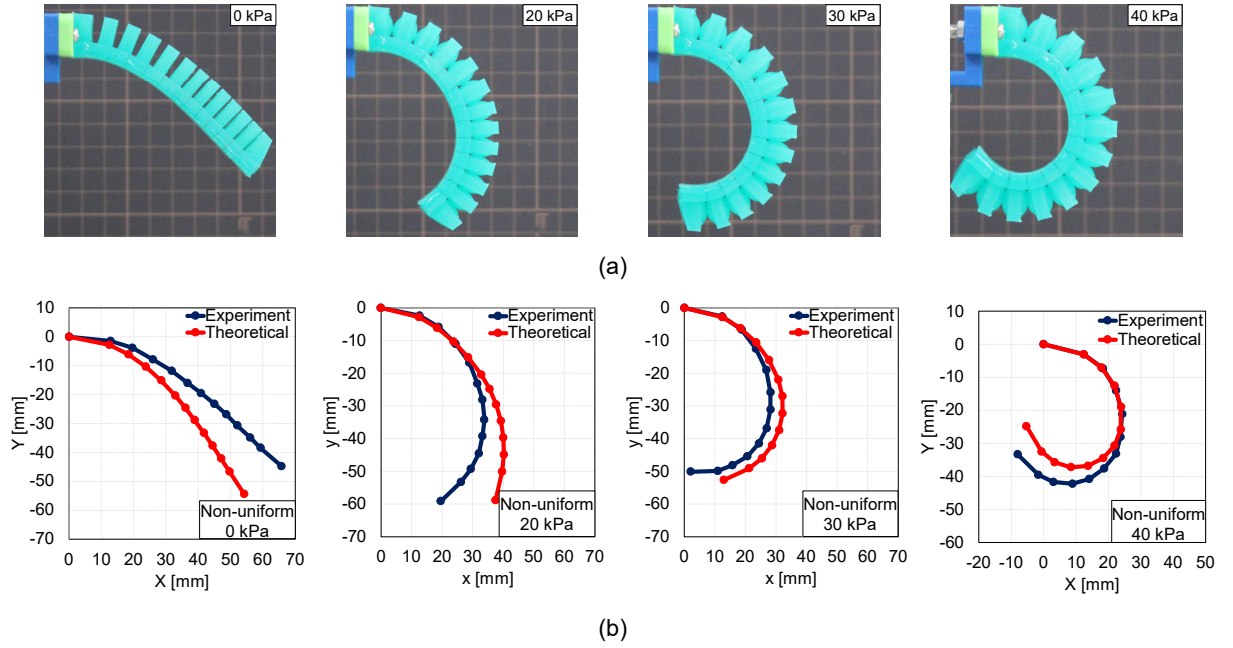


Figure 9: Free space deformation of a nonuniform geometry actuator: (a) snapshots of experimental deformation and (b) plots of experimental and analytical model estimated deformation.

6 Actuators Having Different Length

In order to see, how accurately the analytical model predicts the deformation behavior of actuators having different number of air chambers and having different length. We designed three actuators having different number of air chambers and different length, and carried out the FE and analytical simulations. The actuators have 9, 11, and 13 chambers of identical geometry and have length 66 mm, 78 mm, and 90 mm, respectively. The free space deformation simulation results are presented in Fig. 10, Fig. 11, and Fig. 12. Type C and D actuators were made of Dragon Skin 20 and 30 silicone rubbers, respectively. When compared to the FE simulations, the analytical simulations have the same tendency as they have in Fig. 9 in the main manuscript. The block force simulation results for 0 mm offset are presented in Fig. 13. Based on the observed behavior, we can conclude that the presented model can be applied to actuators having different length.

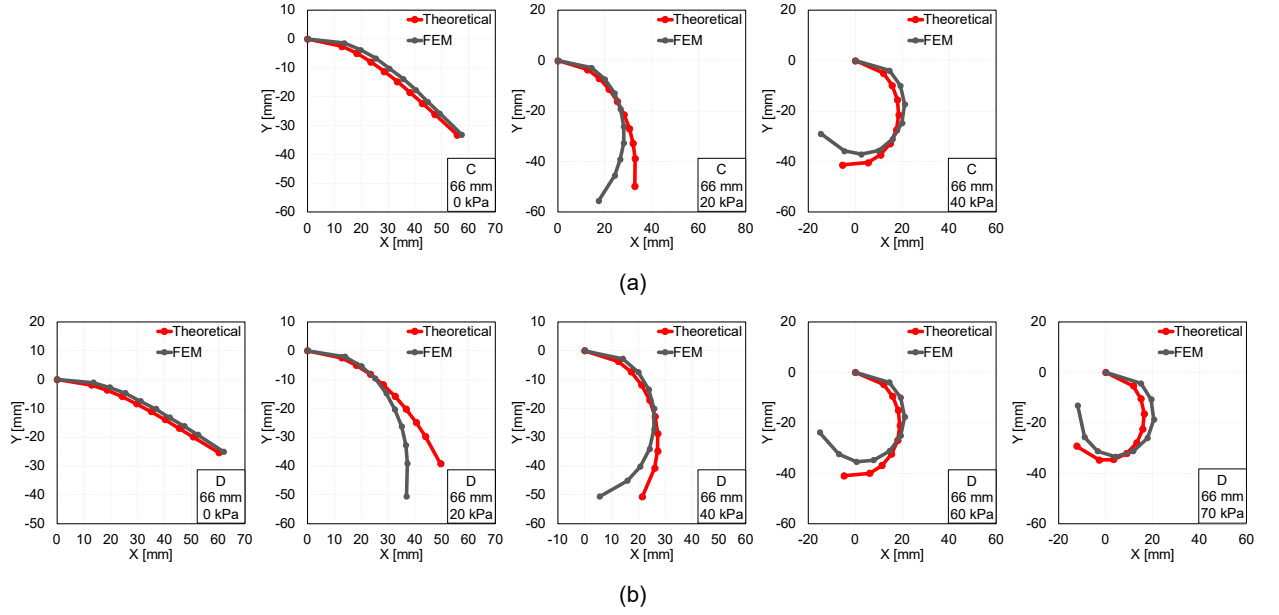


Figure 10: Theoretical and FEM free space deformation plots of 66 mm length actuator.

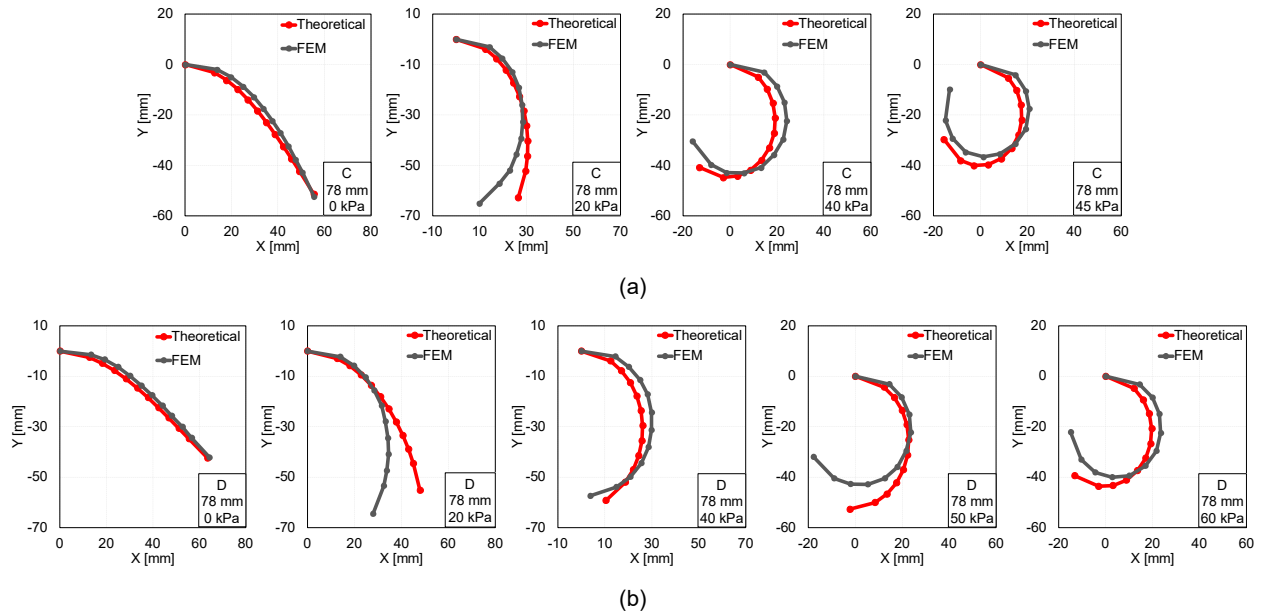


Figure 11: Theoretical and FEM free space deformation plots of 78 mm length actuator.

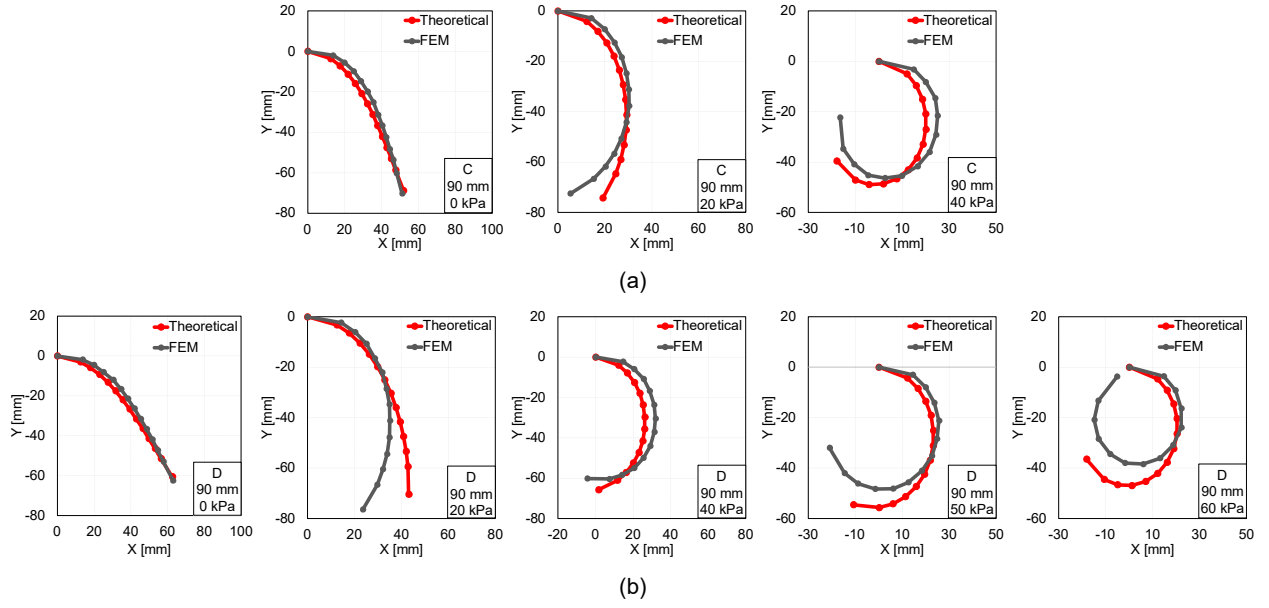


Figure 12: Theoretical and FEM free space deformation plots of 90 mm length actuator.

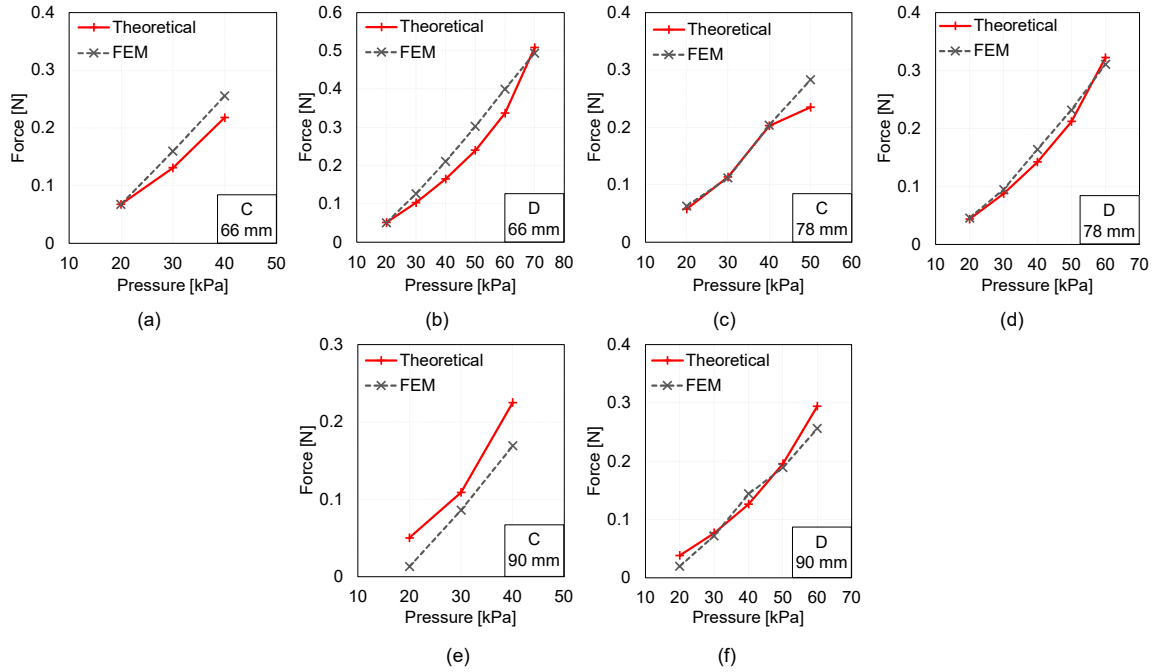


Figure 13: Theoretical and FEM block force plots of (a) type C 66 mm, (b) type D 66 mm, (c) type C 78 mm, (d) type D 78 mm, (e) type C 90 mm, and (f) type D 90 mm actuators.

References

- [1] A. Srivastava and C.-Y. Hui, "Large deformation contact mechanics of long rectangular membranes. i. adhesionless contact," *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, vol. 469, no. 2160, p. 20130424, 2013.
- [2] A. Libai and J. G. Simmonds, *The nonlinear theory of elastic shells*. Cambridge university press, 2005.
- [3] A. Kydonieffs and A. Spencer, "Finite axisymmetric deformations of an initially cylindrical elastic membrane," *The Quarterly Journal of Mechanics and Applied Mathematics*, vol. 22, no. 1, pp. 87–95, 1969.
- [4] R. W. Ogden, *Non-linear elastic deformations*. Dover Publication, 1997.